

Resolved Neutral Hydrogen in an Infalling Galaxy Population: a 100-hour SKA-Mid Pathfinder Program

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Requested telescope: SKA-Mid • Requested time: 100 hours • Array: AA*, Band 2

Abstract

We request 100 hours on SKA-Mid to map the resolved 21-cm HI emission of twelve galaxies in the outskirts and infall region of a nearby, low-mass cluster ($D \approx 20$ Mpc), at $\approx 15''$ resolution and 2.8 km/s velocity resolution. HI is the most extended and least gravitationally bound baryonic component of a galaxy, so it is usually the first thing disturbed or removed as a galaxy falls into a group or cluster: tidal tails, warps, truncated discs and one-sided 'jellyfish' tails all show up in the atomic gas well before the stars are affected. HI surveys with the VLA, WSRT, ASKAP and MeerKAT have found the first examples of this around the nearest, best-studied systems, but tend to run out of either depth or resolution further out. SKA-Mid AA* (144 dishes: 80 new 15-m SKA dishes plus the 64-dish MeerKAT array) reaches a 3σ column density of $\approx 2.1 \times 10^{19} \text{ cm}^{-2}$ at $15'' \times 13''$ resolution in 6 hours per target — deep and sharp enough to trace these features across our whole sample within a single, well-defined observing block. The program is kept deliberately small: a pre-selected sample rather than a blind search, sized to fit one semester and to shake out the SKA-Mid HI pipeline ahead of any larger allocation.

1. Scientific Justification

1.1 HI and environmental processing

Neutral hydrogen is the raw material for star formation, and its distribution and kinematics record how gas is accreted, consumed and expelled over a galaxy's life [1]. Because HI discs extend well beyond the stellar body and are only weakly bound in their outer parts, HI responds to tides, harassment, and the ram pressure of a hot intracluster or intragroup medium long before the stars do [2,3]. Interferometric HI imaging with the WSRT, VLA, ASKAP and MeerKAT has built up a detailed picture of the nearby HI Universe over roughly two decades: THINGS mapped 34 nearby disc galaxies at about $6''$ resolution [4], and the ALFALFA and WALLABY surveys pushed this out to tens of thousands of galaxies, at coarser resolution, using single-dish and interferometric techniques respectively [5,6]. Pointed MeerKAT programs such as MHONGOOSE have since gone deeper on individual targets [7], and these deeper data are what first revealed the tidal tails, bridges, warps and jellyfish morphologies this proposal is designed to follow up [2].

The Fornax cluster is a useful testbed precisely because it is close enough ($D \approx 20$ Mpc) for precursor telescopes to have already found disturbed systems in it. Kleiner et al. used MeerKAT to show that Fornax dwarf galaxies lose their HI rapidly on infall, ahead of any visible change in their stars [8], and Serra et al. resolved the ram-pressure-stripped tail of the interacting galaxy NGC 1427A in detail [9]. Both studies are confined to galaxies near the cluster core, though, where stripping is most advanced and therefore easiest to see. The outskirts and the filament feeding the cluster — where the earliest and faintest stages of gas removal should show up — have not been imaged at comparable depth.

1.2 Why SKA-Mid AA*

SKA-Mid AA* combines 80 new 15-m SKA dishes with the 64-dish MeerKAT array [10,11], and can already deliver the surface-brightness sensitivity and angular resolution this science needs together, which no

existing telescope manages at once. Restricting the array to its dense core (baselines ≤ 2 km) keeps most of the collecting area while giving a synthesized beam of order $15''$ — well matched to the few-kpc scale of tidal tails and stripped gas at $D \approx 20$ Mpc. Resolved imaging of this kind is flagged as one of the main applications of SKA-Mid Band 2 HI observations [2,3], and sits well below the scale of the surveys built around the evolving HI mass function or HI intensity mapping, which need thousands of hours rather than a hundred [12,13].

Because the sample is pre-identified rather than found by a blind search, the science case fits inside a single semester. A hundred hours is enough to address the questions below, and it doubles as a shakedown of RFI excision and multi-session joint deconvolution ahead of any larger HI environmental Key Science Programme [12].

1.3 Key questions

- How much of the HI in these infall galaxies is already disturbed or gone before first pericentric passage, and does its morphology — tail, warp, truncation — point to tides or to ram pressure [2,3]?
- Does the degree of disturbance track a galaxy's position in the group/filament/cluster structure, separately from its mass [2,14]?
- For the galaxies with clean, regular discs, do the outer, dark-matter-dominated parts of their rotation curves look different from field galaxies of similar mass [15]?

2. Immediate Objective and Target Sample

The objective is resolved HI imaging, well beyond the depth of existing precursor data, of 12 galaxies spread across the outskirts, filament and infall region of the Fornax cluster ($D \approx 20$ Mpc). The sample is drawn from existing MeerKAT and ASKAP HI catalogues and from the targeted studies cited above [7,8,9], so every pointing already has a known target rather than relying on a blind search. It spans a range of projected clustercentric distance — from just inside the virial radius out into the connecting filament — and a mix of regular, mildly disturbed and confirmed tailed or jellyfish systems, so that trends with environment and mass can be read off a single, uniformly observed dataset.

Each target gets one pointing. At Band 2 the SKA1 15-m primary beam is about a degree across, comfortably containing both the stellar disc and any surrounding tidal or stripped HI, which typically extends tens of kpc [2].

3. Description of the Observations

3.1 Array configuration and spectral setup

We request SKA-Mid AA*, Band 2 (0.95–1.76 GHz), tuned to the group's recession velocity (illustrative centre 1413.3 MHz, $cz \approx 1500$ km/s). A 4 MHz zoom window (≈ 850 km/s) centred on each target's systemic velocity covers the full line plus enough baseline either side for continuum subtraction. Native 13.4 kHz channels (≈ 2.8 km/s) resolve non-circular motions in the disturbed gas; we smooth to ≈ 8 km/s in post-processing to bring out faint, extended emission. Robust weighting (robust = 0) on the core sub-array (baselines ≤ 2 km) gives a synthesized beam of $\approx 15.2'' \times 12.6''$, matched to the few-kpc scale of the features we are trying to resolve. Table 1 gives the full setup and time request; Table 2 gives the resulting sensitivity, both from the SKAO Sensitivity Calculator [16].

Parameter	Value
Telescope / array	SKA-Mid, Array Assembly * (AA*); 144 dishes (80 SKA-Mid 15-m + 64 MeerKAT 13.5-m), core sub-array restricted to baselines ≤ 2 km

Parameter	Value
Receiver band	Band 2 (0.95–1.76 GHz)
Tuning	Centred at 1413.3 MHz ($cz \approx 1500$ km/s, matched to the Fornax infall region)
Spectral setup	Zoom window, 4 MHz bandwidth (≈ 850 km/s), 13.4 kHz channels (≈ 2.8 km/s)
Weighting / resolution	Robust (robust = 0), core-restricted array $\rightarrow$ synthesized beam $\approx 15.2'' \times 12.6''$
Targets	12 group/cluster-infall galaxies in the Fornax outskirts and associated filament
On-source time per target	6.0 h
Total on-source time	72.0 h
Calibration & overheads	≈ 22.0 h (bandpass/flux/phase calibration, pointing, slew, RFI margin)
Total time requested	94.0 h (rounded to 100 h)

3.2 Sensitivity

Six hours on-source per target gives a native-channel (2.8 km/s) rms of $254 \mu\text{Jy}/\text{beam}$ at this resolution. Smoothed to 8.3 km/s — a three-channel average matched to the linewidths expected in disturbed, extraplanar HI — this is a 3σ column density limit of $\approx 2.1 \times 10^{19} \text{ cm}^{-2}$. That is roughly in line with the deepest pointed MeerKAT surveys of similar targets [7], but reached here at higher angular resolution and with 53 fewer dishes than the full AA4 array will eventually provide. For an unresolved clump with a 20 km/s line, the 5σ mass limit at $D = 20$ Mpc is $\approx 9 \times 10^5 M_\odot$: low enough to pick up isolated HI debris stripped from the main discs, not just the discs themselves. These are calculator estimates for average weather and elevation; actual depths will vary with conditions and RFI on the night, and would improve further once the array grows toward AA4.

Quantity	Value
Native channel rms (2.8 km/s, 6 h)	$254 \mu\text{Jy}/\text{beam}$
Smoothed rms (8.3 km/s, 3-channel average)	$\approx 147 \mu\text{Jy}/\text{beam}$
3σ HI column density ($15.2'' \times 12.6''$ beam, 8.3 km/s)	$\approx 2.1 \times 10^{19} \text{ cm}^{-2}$
5σ integrated HI mass limit (unresolved, 20 km/s line, $D = 20$ Mpc)	$\approx 9 \times 10^5 M_\odot$
Surface-brightness sensitivity (natural units)	$\approx 0.81 \text{ K (per native channel)}$

3.3 Time justification

Twelve targets at 6 hours each is 72 hours on-source. We add about 22 hours ($\approx 30\%$) for calibration, pointing, slewing and RFI margin — Band 2 spectral-line data routinely lose channels to interference even at SKA's radio-quiet sites [12] — for a total of 94 hours, rounded up to 100 for contingency. Six hours is close to the least time that resolves the diagnostic scale here (a few kpc, tens of arcsec) at a useful depth; anything shorter would only recover the bright disc emission already seen by precursor surveys, not the faint tails and warps that motivate this proposal.

4. Data Reduction and Analysis Plan

- Calibration (bandpass, flux, phase), RFI flagging and joint deconvolution will use the standard SKA Science Data Processor pipeline [10,12].
- HI moment maps — integrated intensity, velocity field, velocity dispersion — will be made at both native and smoothed resolution.
- Each galaxy will be classified by HI morphology (disc, warp, truncation, tail) and compared against clustercentric distance and mass, following the diagnostics used in precursor cluster and group HI work [2,3].
- Rotation curves will be extracted for the subset with regular kinematics and compared with mass-matched field galaxies [15].
- Calibrated visibilities and image products will go into the SKA Regional Centre Network, per SKAO data policy [17].

5. Feasibility and Risk

- RFI: Band 2 carries known satellite and ground-based interference at some frequencies. The tuning avoids the worst of it at this redshift, and the 30% overhead allows for extra flagging [12].
- Scheduling: HI spectral-line work is best done at night to avoid solar interference; this is built into the overhead and does not otherwise limit visibility for these targets.
- Array readiness: this program only needs AA* (144 dishes), and within that, only the already-commissioned core (baselines ≤ 2 km) — so it does not depend on the outer spiral arms being finished. If the array grows toward AA4 before this is scheduled, the same targets would simply be reached at greater depth in the same time.

References

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